

Project: Resilient and Scalable Web Application Deployment on AWS

Implementation & Configuration Guide

- **Date:** August 2026
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- **Project Title:** Resilient and Scalable Web Application Deployment on AWS

1. Introduction & Scope

- **Purpose:** To design, implement, and document a highly available, fault-tolerant, and auto-scaling web application architecture on AWS.
- **Scope:** Covers VPC networking across multiple Availability Zones, automated compute scaling, load balancing, shared file storage, and DNS management.

2. Architecture Overview

- The architecture is deployed in the ap-south-1 (Mumbai) region and utilizes a custom Virtual Private Cloud (VPC) to ensure network isolation and security.
- **High Availability:** Resources are distributed across multiple Availability Zones (ap-south-1a and ap-south-1b).
- **Traffic Flow:** Internet users access the application via domain name (milanrai.me) managed by Route 53, which routes traffic to an Application Load Balancer (ALB). The ALB securely distributes incoming requests to healthy EC2 instances running inside the Auto Scaling Group.

3. AWS Services Configuration

- **Virtual Private Cloud (VPC):**
 - CIDR Block: 192.168.0.0/24
 - Public Subnets (192.168.0.0/26, 192.168.0.64/26) for ALB and NAT Gateway.
 - Private Subnets (192.168.0.128/26, 192.168.0.192/26) for secure EC2 hosting.
- **Elastic Compute Cloud (EC2) & Auto Scaling:**
 - Configured an Auto Scaling Group (ASG) with a minimum capacity of 2 and maximum capacity of 5 instances.
 - Target tracking scaling policies set up to monitor CPU utilization.
- **Elastic File System (EFS):**
 - Mounted across EC2 instances to maintain persistent, shared file storage.
- **Application Load Balancer (ALB):**
 - Listens on port 80/8080 and routes traffic to target groups across multi-AZs.

- **Route 53:**
 - Configured DNS records (A-record Alias pointing to the ALB) for the domain milanrai.me.

4. Application Deployment & Security

- **Deployment:** Deployed a Python Flask web application paired with a real-time telemetry dashboard using a custom startup script (User Data).
- **Security Groups:** Restricted inbound traffic to allow HTTP (port 80) and application traffic (port 8080) exclusively from the ALB security group to the web servers.

Performance and Optimization Report

- **Load Testing:** Performed stress testing using the stress utility combined with Python's multiprocessing module to simulate 100% CPU loads.
- **Scaling Behavior:** Successfully triggered the CloudWatch alarms and Auto Scaling triggers, scaling out new instances dynamically and scaling in when the load normalized.